

# Rittik Sharma

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## EDUCATION

**Indian Institute of Technology Patna**  
*B.Tech in Electrical and Electronics Engineering*  
**DAV Public School, Cantt, Gaya, Bihar**  
*Class XII, CBSE*  
**DAV Public School, Kaiya, Gaya, Bihar**  
*Class X, CBSE*

Patna, Bihar, India  
Nov 2022 - May 2026  
May 2021  
Grade: **95.4%**  
May 2019  
Grade: **96.0%**

## EXPERIENCE

**Vivriti Capital Pvt. Ltd.** Chennai, Tamil Nadu  
*Software Development Intern - Research & Development*  
Jan 2026 – June-2026

- **Designed** a full-stack **Data Vault Management System** utilizing **Spring Boot**, **React.js**, and **REST APIs**, establishing secure document storage and enterprise-level workflow management.
- **Architected** scalable backend modules for secure file upload, metadata processing, and lifecycle management, reducing **data retrieval latency**.
- **Benchmarked** and analyzed **Ory Hydra** alongside **Keycloak** to evaluate **OAuth2/OpenID** authentication capabilities, scalability, and robust token management.
- **Implemented** a customized **IAM architecture** integrating **Ory Kratos** and **Keto** to enforce centralized **Role-Based Access Control (RBAC)**, secure session management, and API security.
- **Constructed** a high-performance **C++ File Compression System** supporting **Text, PDF, MP4, and WebM** formats, optimizing encoding workflows and significantly reducing storage overhead.

## PROJECTS

**File Compression Tool** | C++, FFmpeg, Ghostscript, Zstd | March 2026 [Source Code](#)

- **Engineered** a cross-platform CLI tool supporting **4+ file formats** with **100% offline execution** capability across Windows and Linux environments.
- **Integrated** 3 distinct compression engines (**Zstd, FFmpeg, Ghostscript**), successfully achieving a **60–80% reduction** in overall file size.
- **Formulated** a **4-level compression hierarchy** featuring custom bitrate and DPI tuning, delivering a **30–50% improvement** in size-to-quality ratio control.
- **Developed** an **extension-independent** file signature detection system, boosting processing accuracy and reliability by approximately **40%**.

**ResumeIQ Hiring Assistant** | Next.js, TypeScript, NestJS | May 2026 [Source Code](#)

- **Built** a full-stack candidate screening web app using **Next.js** (SSG/SSR) and **TypeScript**, connected to **NestJS** REST APIs and a **MySQL** database to ingest and index resumes.
- **Implemented** a resume ingestion pipeline that processed resumes across **5+ sections** (skills, experience, education, projects, certifications), extracting and normalizing candidate data into a structured MySQL schema.
- **Engineered** an **AI/LLM-assisted** matching pipeline using RAG with a vector store (FAISS) to improve candidate-job matching precision by **15%** on internal tests.
- **Built** a containerized **4-microservice** ecosystem using **Docker**, reducing environment configuration issues and accelerating local development and testing workflows.

## ACHIEVEMENTS

- **Attained** a 4-Star rating on **CodeChef** (Peak Rating: **1808**) and a **1564** rating on **Codeforces**, demonstrating consistent algorithmic performance. [Profile link](#) | [Profile link](#)
- **Mastered** advanced **Data Structures and Algorithms** by successfully solving and optimizing **500+ complex problems** across tier-one coding platforms.
- **Ranked 62nd globally** in CodeChef Starters 128 (Div 3) among **4,400+** participants, and **90th globally** in Starters 173 (Div 2) out of **5,000+** competitors.
- **Secured a global rank of 479** out of **25,000+ competitors** in Educational Codeforces Round 174.
- **Achieved top-percentile academic standing** with an All India Rank (AIR) of **5102** in JEE Advanced 2022 and **7911** in JEE Mains 2022.

## SKILLS

**Languages:** C, C++, Kotlin, Java, JavaScript, Python, SQL  
**Core:** DSA, OOP, DBMS, OS, Computer Networks, System Design, REST APIs  
**Tools / Frameworks:** HTML, CSS, Node.js, Express.js, Spring Boot, ReactJS, Docker, Git, GitHub, Bitbucket, Jira, Confluence, JUnit 5, MySQL, MongoDB, Linux, IntelliJ IDEA, VS Code, Postman, Spinnaker

## EXTRA-CURRICULAR

- Served as the **Coordinator** of the esports unit for **IIT Patna's annual sports fest Infinito**, managing all teams and ensuring smooth, conflict-free event execution. 2024